

Provider-Service Billing Patterns and Audit Targeting in Medicare Professional Claims

Giorgos Tzimas

July 15, 2026

Abstract

Payment-integrity work often begins with a practical question: which records should be reviewed first when the number of possible claims, providers, codes, and policies is too large for manual review? This project studied whether transparent provider-service features from public Medicare professional payment data can concentrate review attention on unusual billing patterns without treating an algorithmic score as proof of improper billing. The analysis used the CMS Medicare Physician and Other Practitioners by Provider and Service dataset for selected high-volume HCPCS codes in Indiana and Illinois during 2023 and 2024 [5, 6]. The scoring logic combined SQL-derived utilization and payment features with percentile-based indicators for payment volume, service intensity, charge-to-allowed ratios, payment per beneficiary, beneficiary-day intensity, and year-over-year provider-code growth.

The analytic file contained 237,405 provider-service records. In 2024, the selected cohort included 51,101 providers and an estimated \$1.18 billion in Medicare payments. The highest audit-priority decile represented 10.0% of 2024 records but captured 30.7% of estimated Medicare payments and 31.7% of services. These findings support the hypothesis that transparent, claims-derived features can materially improve audit targeting efficiency, while also showing why follow-up review must include coding policy, documentation, and clinical context.

1 Introduction

Healthcare claims data are dense, administrative, and useful. They are also easy to misuse if analytic outputs are treated as conclusions rather than leads. Professional claims encode who rendered a service, which HCPCS or CPT-related code was billed, where the service occurred, how often it was provided, how many beneficiaries were represented, and how submitted charges compared with Medicare allowed and paid amounts [5, 1]. That structure makes the data valuable for compliance monitoring, utilization review, and payment-integrity analytics.

The central operational challenge is not only finding large payments. A useful audit method should identify patterns that are unusual relative to peers, codes, places of service, and time. A provider-service record with high total payment may simply reflect high volume in a common service. A record with high service intensity, high charge-to-allowed ratio, or rapid year-over-year growth may be more relevant to a focused review if the pattern is unusual for the same HCPCS code and setting. CMS coding and billing resources, including NCCI edits, HCPCS guidance, physician fee schedule materials, and Internet-only manuals, reflect how much payment integrity depends on the translation of policy and coding rules into operational logic [3, 4, 2].

Prior work on anomaly detection and health care fraud detection emphasizes the same caution. Unusual observations are not automatically fraudulent, especially in domains where labels are rare, behavior is heterogeneous, and legitimate clinical variation is common [7, 9]. More recent work also argues that

payment-integrity models need explainability so that reviewers understand why a provider or claim was surfaced [10, 8]. This project follows that practical framing. The goal is not to infer fraud. The goal is to build an interpretable record-selection score that can support review prioritization.

2 Research Question and Hypothesis

The primary research question was whether transparent provider-service features among selected high-volume Medicare professional HCPCS services in Indiana and Illinois could identify a small review cohort that captures a disproportionate share of estimated Medicare payments and service volume.

The primary hypothesis was that a composite audit-priority score based on payment volume, service intensity, charge-to-allowed ratios, payment per beneficiary, beneficiary-day intensity, and year-over-year provider-code growth would concentrate estimated Medicare payment in the highest-risk decile beyond the 10 percent of records selected. A secondary descriptive question examined which service families and provider specialties contributed most to flags for high service intensity, high charge ratios, and rapid growth.

3 Data

The analysis used the CMS Medicare Physician and Other Practitioners by Provider and Service data, which are organized by NPI, provider type, HCPCS code, and place of service [5]. The extract covered 2023 and 2024 records for Indiana and Illinois. The HCPCS cohort included common evaluation and management codes, emergency department E/M, subsequent hospital care, electrocardiogram, and therapeutic procedure code families: 99203, 99204, 99213, 99214, 99215, 99232, 99285, 93000, and 97110.

This code set was selected because it provides a compact, policy-relevant mix of office visits, hospital care, emergency department care, diagnostic testing, and therapy services. It also keeps the project focused enough for clear review logic. The dataset is public and aggregated at the provider-service level. It does not contain patient-level claim lines, diagnosis codes, modifiers, medical record text, or adjudication outcomes. Those omissions are important because coding compliance ultimately depends on documentation and detailed claim context [12, 11].

Table 1: Downloaded CMS provider-service records by year and state

Year	State	HCPCS codes	Records
2023	IL	9	76,672
2023	IN	9	41,229
2024	IL	9	78,037
2024	IN	9	41,467

4 Methods

4.1 Data Processing

The pipeline downloaded filtered CMS API extracts by year, state, and HCPCS code, then standardized field names and numeric types. Core fields included rendering NPI, provider type, provider state, HCPCS code, HCPCS description, place of service, total beneficiaries, total services, beneficiary-day services, average submitted charge, average Medicare allowed amount, average Medicare payment, and average Medicare standardized amount.

Derived totals were estimated by multiplying the average dollar amounts by total services:

$$\text{estimated submitted charge} = \text{total services} \times \text{average submitted charge}$$
$$\text{estimated allowed amount} = \text{total services} \times \text{average Medicare allowed amount}$$
$$\text{estimated Medicare payment} = \text{total services} \times \text{average Medicare payment}$$

The project also created service-intensity and payment-ratio features:

$$\text{services per beneficiary} = \frac{\text{total services}}{\text{total beneficiaries}}$$
$$\text{payment per beneficiary} = \frac{\text{estimated Medicare payment}}{\text{total beneficiaries}}$$
$$\text{charge-to-allowed ratio} = \frac{\text{average submitted charge}}{\text{average Medicare allowed amount}}$$

A SQLite database was created for review-style analysis. SQL views summarized yearly state totals, provider payment profiles, provider-type and HCPCS summaries, and high-volume code summaries. This structure mirrors how claims review logic is often maintained: raw extracts are preserved, derived fields are documented, and downstream summaries can be rerun from stable tables.

4.2 Audit-Priority Score

Within each year, state, HCPCS code, and place-of-service group, the pipeline ranked records by estimated Medicare payment, services per beneficiary, charge-to-allowed ratio, payment per beneficiary, and beneficiary-day intensity. The score used percentile ranks rather than raw values so that records were compared against peers billing the same code in the same general setting.

The score also incorporated a year-over-year growth flag for provider-HCPCS combinations observed in both years. A 2024 record was flagged for rapid growth when both services and estimated Medicare payment increased by more than 50 percent compared with 2023 and the 2024 provider-code payment exceeded the median provider-code payment. This is a practical screen for focused review, not a claim that growth is inappropriate.

The final score ranged approximately from 0 to 100. Higher scores reflected more indicators that the provider-service record was unusual relative to similar records. Records were then divided into annual deciles, with decile 10 representing the highest audit-priority group.

4.3 Statistical Analysis

The primary analysis was descriptive and operational. The main outcome was payment capture: the share of estimated Medicare payments captured by each audit-priority decile. The hypothesis was supported if the top decile captured much more than 10 percent of 2024 estimated Medicare payments. Secondary analyses summarized service capture, flag counts, service families, and provider types. No record was labeled as fraud or improper billing. The score is best interpreted as a prioritization tool for review teams that would still need coding rules, medical records, documentation, and policy context.

5 Results

5.1 Analytic Cohort

The final analytic dataset included 237,405 provider-service records across 2023 and 2024. In 2024, the selected services represented 51,101 unique providers and \$1.18 billion in estimated Medicare payments. Estimated payment in the selected cohort decreased by 2.0% from 2023 to 2024.

Table 2: 2024 provider-service cohort by code family

Code family	Records	Providers	Services	Payment, \$M
Established outpatient E/M	70,722	35,767	10,592,323	794.0
Subsequent hospital care	12,000	11,957	2,243,133	134.2
New outpatient E/M	23,807	17,496	1,102,143	102.5
Emergency department E/M	4,088	4,088	633,701	83.6
Therapeutic procedure	5,684	5,683	3,768,222	64.0
Electrocardiogram	3,203	3,203	381,874	3.8

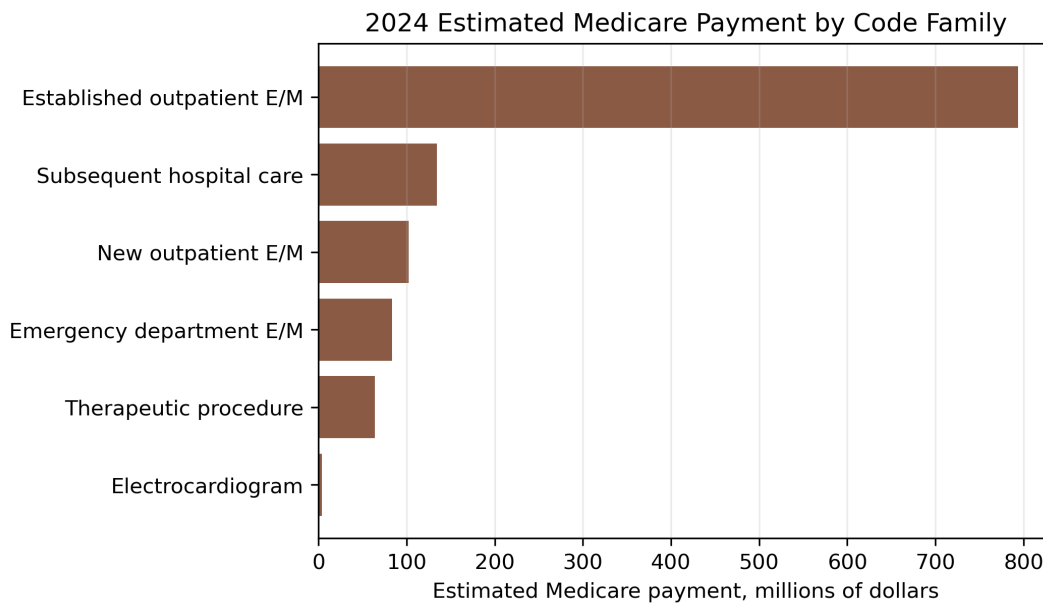


Figure 1: Estimated 2024 Medicare payment by selected HCPCS code family.

5.2 Audit Score Distribution

Audit-priority scores varied widely across the selected provider-service records. The score distribution was right-skewed, which is expected when several percentile-based indicators are combined. Many records had moderate payment and intensity profiles, while a smaller group had several review indicators at once.

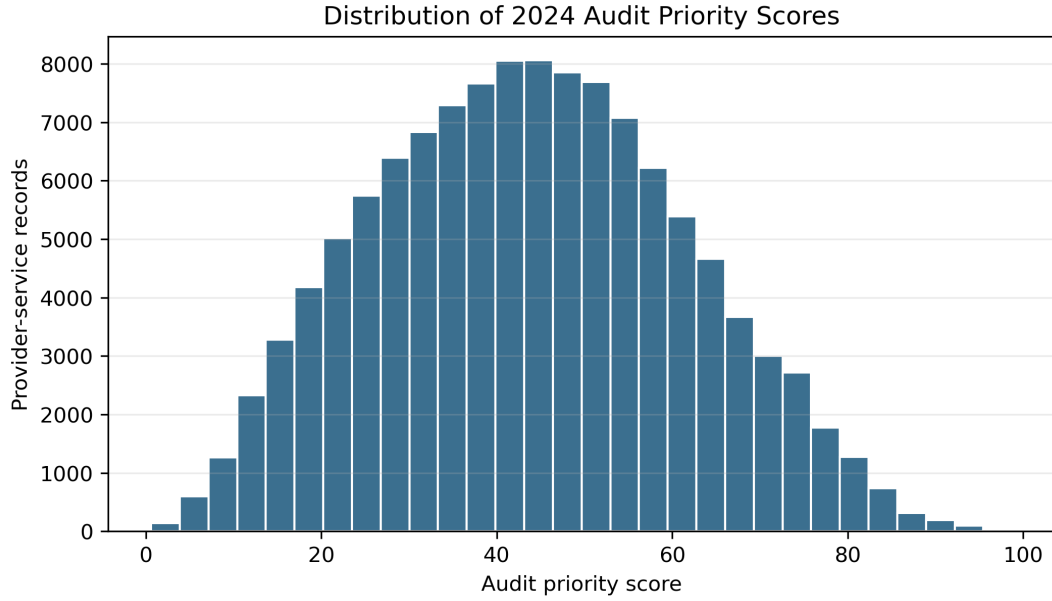


Figure 2: Distribution of audit-priority scores among 2024 provider-service records.

5.3 Payment Capture by Review Decile

The top audit-priority decile contained 11,951 2024 provider-service records, or 10.0% of records. It captured 30.7% of estimated Medicare payments and 31.7% of services. This is roughly three times the payment share expected from an untargeted 10 percent sample.

Table 3: 2024 audit-priority decile summary

Decile	Records	Payment share	Service share	Charge flags	Intensity flags
1	11,950	1.6%	1.9%	26	0
2	11,950	2.6%	2.8%	803	0
3	11,951	3.6%	3.8%	864	0
4	11,950	4.7%	4.8%	1,036	1
5	11,951	6.0%	6.1%	1,176	3
6	11,950	7.5%	7.4%	1,173	6
7	11,950	10.3%	9.7%	1,259	64
8	11,951	13.6%	13.1%	1,503	353
9	11,950	19.3%	18.8%	1,760	1,430
10	11,951	30.7%	31.7%	2,368	7,249

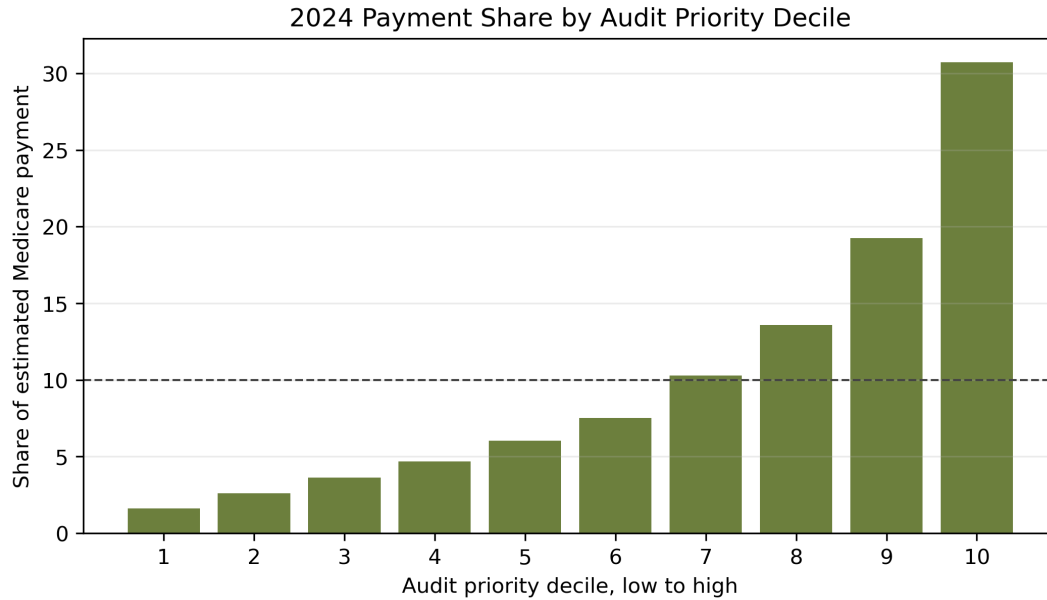


Figure 3: Share of estimated Medicare payment captured by each 2024 audit-priority decile.

5.4 Provider-Type Patterns

Provider types with the largest estimated payments were not always the same as those with the highest number of flags. Internal medicine, nurse practitioners, family practice, emergency medicine, cardiology, physical therapists in private practice, and physician assistants were prominent in the selected cohort. In the top decile, the score surfaced 2,368 high charge-ratio flags, 7,249 high service-intensity flags, and 2,420 rapid growth flags.

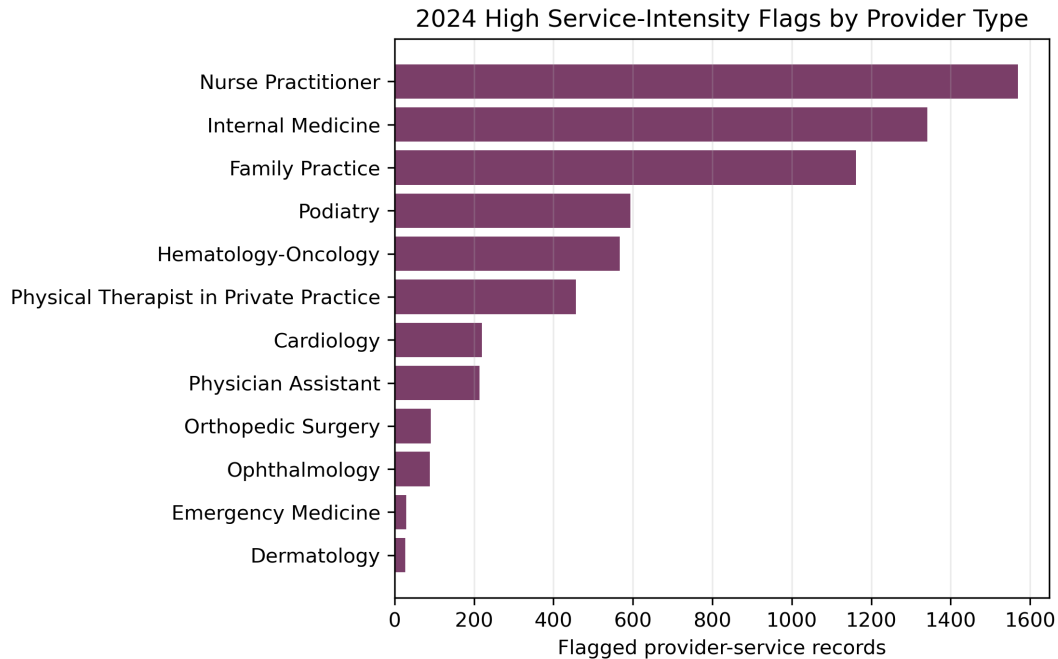


Figure 4: High service-intensity flags by provider type in 2024.

5.5 Year-over-Year Growth

Growth flags were designed to catch provider-HCPCS combinations with both service growth and payment growth. These patterns may reflect changes in practice mix, coding updates, access shifts, or billing behavior. They are useful for review selection because they help teams find records where the current-year pattern changed materially from the prior year.

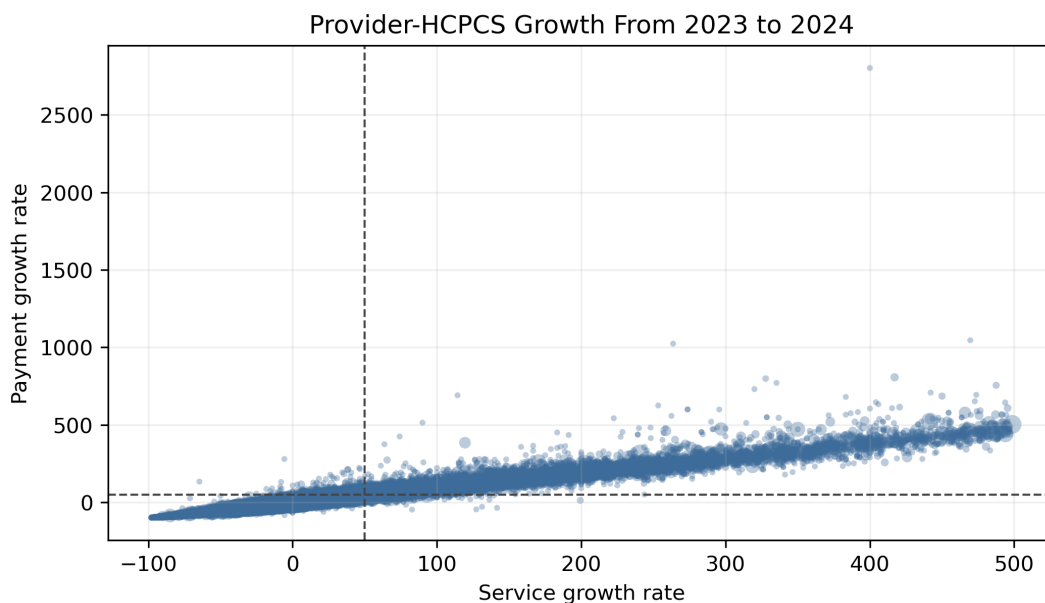


Figure 5: Provider-HCPCS service and payment growth from 2023 to 2024.

6 Discussion

The findings support the primary hypothesis. A transparent audit-priority score based on provider-service features concentrated 2024 estimated Medicare payments into a small review cohort. The top decile captured 30.7% of estimated payment while representing only 10.0% of records. That is the core operational value of the approach: it gives review teams a defensible starting point when time and review resources are limited.

The score is intentionally simple. This is a strength for audit operations because each component can be explained: high payment volume, high services per beneficiary, high charge-to-allowed ratio, high payment per beneficiary, high beneficiary-day intensity, and rapid growth. A reviewer can trace why a record was selected. That matters in compliance settings, where black-box predictions are less useful than rules and features that can be discussed with coding, clinical, and audit staff [10].

The results also show why payment-integrity analytics should not stop at payment totals. Some high-payment services are common and clinically expected. Other records combine high payment with intensity, ratio, or growth indicators. A composite score can rank records in a way that is closer to review workflow than a single dollar threshold. This is consistent with broader anomaly-detection guidance, where local context and feature design are central to the usefulness of a flagged observation [7].

7 Limitations

This analysis used public provider-service aggregates rather than patient-level claims. It did not include ICD-10 diagnoses, modifiers, claim line dates, denial outcomes, medical record documentation, payer edits, or clinical notes. As a result, the score cannot determine whether any billing pattern is compliant or noncompliant. It can only identify records that may deserve review.

The HCPCS cohort was deliberately limited to selected high-volume services in Indiana and Illinois. Results should not be generalized to all Medicare professional services, Medicaid claims, inpatient claims, or commercial insurance data. The score also depends on percentile thresholds and weights chosen for interpretability. In a production setting, those weights should be reviewed with coding, clinical, and audit stakeholders and monitored over time.

Finally, submitted charges and Medicare payments are affected by fee schedules, geography, place of service, provider mix, and coding policy. High charge-to-allowed ratios are useful signals for review prioritization, but they are not evidence of improper billing on their own [4, 12].

8 Conclusion

A clear SQL and Python workflow using public CMS provider-service data can produce an interpretable audit-priority score for professional claims review. In this selected HCPCS cohort, the top 10 percent of 2024 records captured about one-third of estimated Medicare payments and services. The project shows how claims features, coding concepts, data validation, and readable rule logic can be combined into a practical payment-integrity targeting method. The score should be used as a triage tool, not as an allegation. Its value is in focusing expert review on records where the data pattern is unusual enough to justify a closer look.

References

- [1] Centers for Medicare & Medicaid Services. Healthcare common procedure coding system, 2026. URL <https://www.cms.gov/medicare/coding-billing/healthcare-common-procedure-system>.
- [2] Centers for Medicare & Medicaid Services. Internet-only manuals, 2026. URL <https://www.cms.gov/medicare/regulations-guidance/manuals/internet-only-manuals-ioms>.

- [3] Centers for Medicare & Medicaid Services. National correct coding initiative edits, 2026. URL <https://www.cms.gov/medicare/coding-billing/national-correct-coding-initiative-ncci-edits>.
- [4] Centers for Medicare & Medicaid Services. Physician fee schedule, 2026. URL <https://www.cms.gov/medicare/payment/fee-schedules/physician>.
- [5] Centers for Medicare & Medicaid Services. Medicare physician & other practitioners - by provider and service, 2026. URL <https://data.cms.gov/provider-summary-by-type-of-service/medicare-physician-other-practitioners/medicare-physician-other-practitioners-by-provider-and-service>. Updated May 21, 2026.
- [6] Centers for Medicare & Medicaid Services. Medicare physician & other practitioners by provider and service data dictionary, 2026. URL <https://data.cms.gov/resources/medicare-physician-other-practitioners-by-provider-and-service-data-dictionary>.
- [7] Varun Chandola, Arindam Banerjee, and Vipin Kumar. Anomaly detection. *ACM Computing Surveys*, 41(3): 1–58, 2009. doi: 10.1145/1541880.1541882.
- [8] Justin M. Johnson and Taghi M. Khoshgoftaar. Medicare fraud detection using neural networks. *Journal of Big Data*, 6(1), 2019. doi: 10.1186/s40537-019-0225-0.
- [9] Hossein Joudaki, Arash Rashidian, Behrouz Minaei-Bidgoli, Mahmood Mahmoodi, Bijan Geraili, Mahdi Nasiri, and Mohammad Arab. Using data mining to detect health care fraud and abuse: A review of literature. *Global Journal of Health Science*, 7(1), 2015. doi: 10.5539/gjhs.v7n1p194.
- [10] Shubhranshu Shekhar, Jetson Leder-Luis, and Leman Akoglu. Unsupervised machine learning for explainable health care fraud detection. Technical Report 30946, National Bureau of Economic Research, 2023.
- [11] U.S. Department of Health and Human Services, Office of Inspector General. Compliance guidance, 2026. URL <https://oig.hhs.gov/compliance/compliance-guidance/>.
- [12] U.S. Department of Health and Human Services, Office of Inspector General. Physician education training materials, 2026. URL <https://oig.hhs.gov/compliance/physician-education/>.